

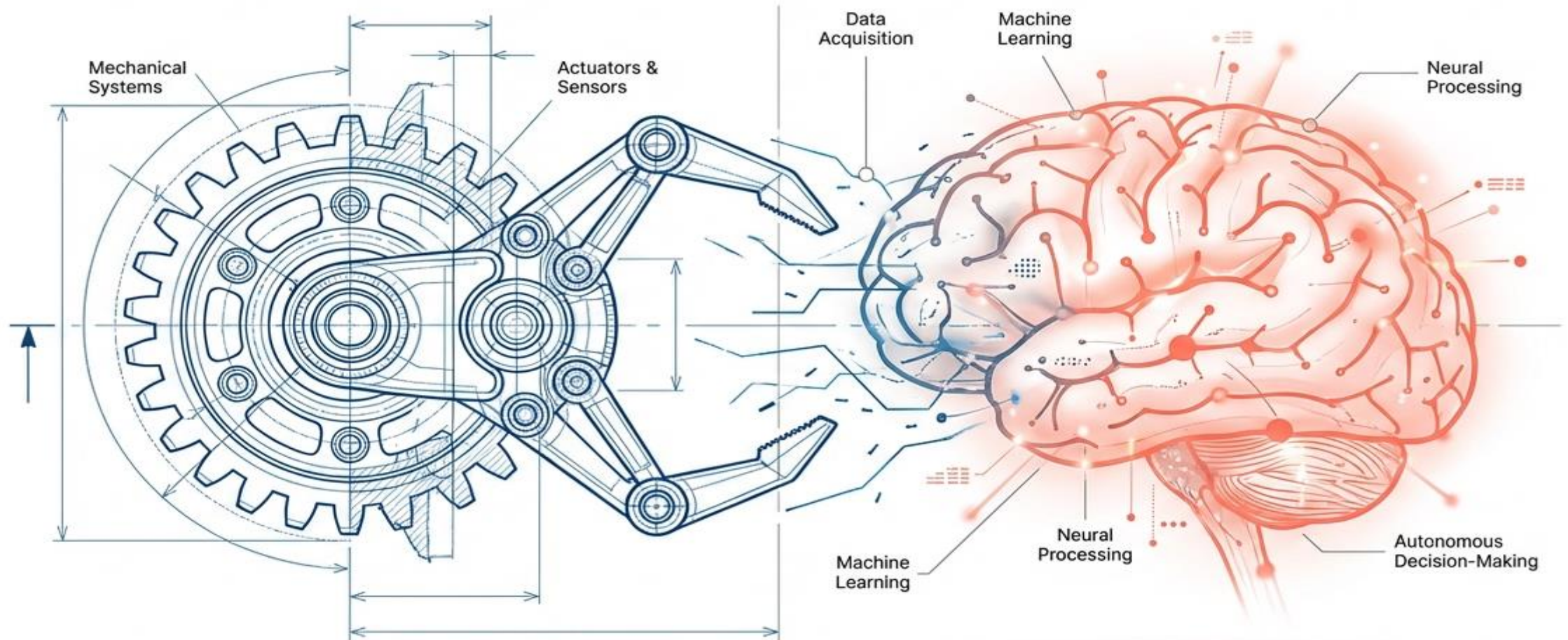


RAI 811: ROBOT MECHANICS & CONTROL



Robot Mechanics and Control: From Automation to Intelligence

An evolutionary overview of definitions, anatomy, and the paradigm shift toward autonomous partnership.





What is a Robot?

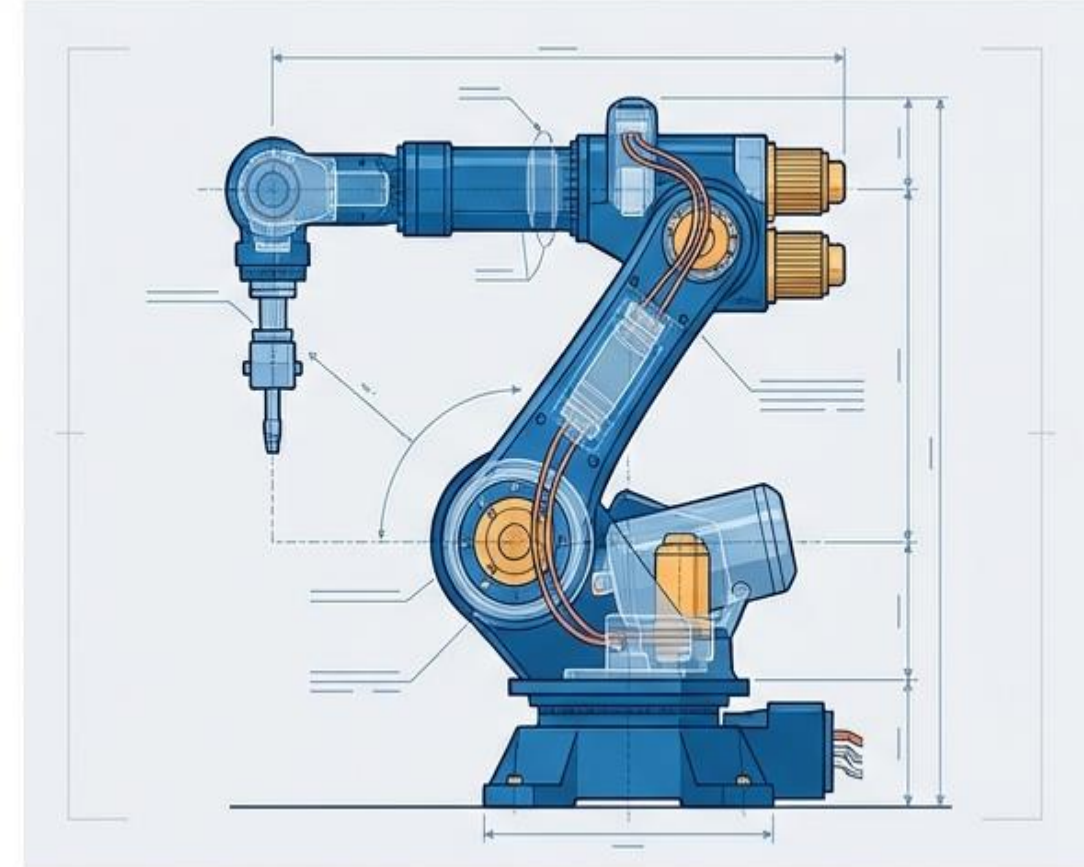
- There is no standard definition of a Robot.
- Joseph Engelberger, who is considered as the Father of Robotics, famously said:
"I can't define a robot, but I know one when I see one."
- As the field of Robotics keeps progressing, new definitions keep coming up.
- Here is a definition that I like:
"Any automatically operated machine that replaces human effort, though it may not resemble human beings in appearance or perform functions in a humanlike manner."

(Encyclopedia Britannica)

What is a Robot?

- **Random House Dictionary:** A machine that resembles a human being and does mechanical routine tasks on command.
- **Robotics Association of America:** An industrial robot is a re-programmable, multifunctional manipulator designed to move materials, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks.

The Industrial Standard



Source: Robotics Association of America (RIA)

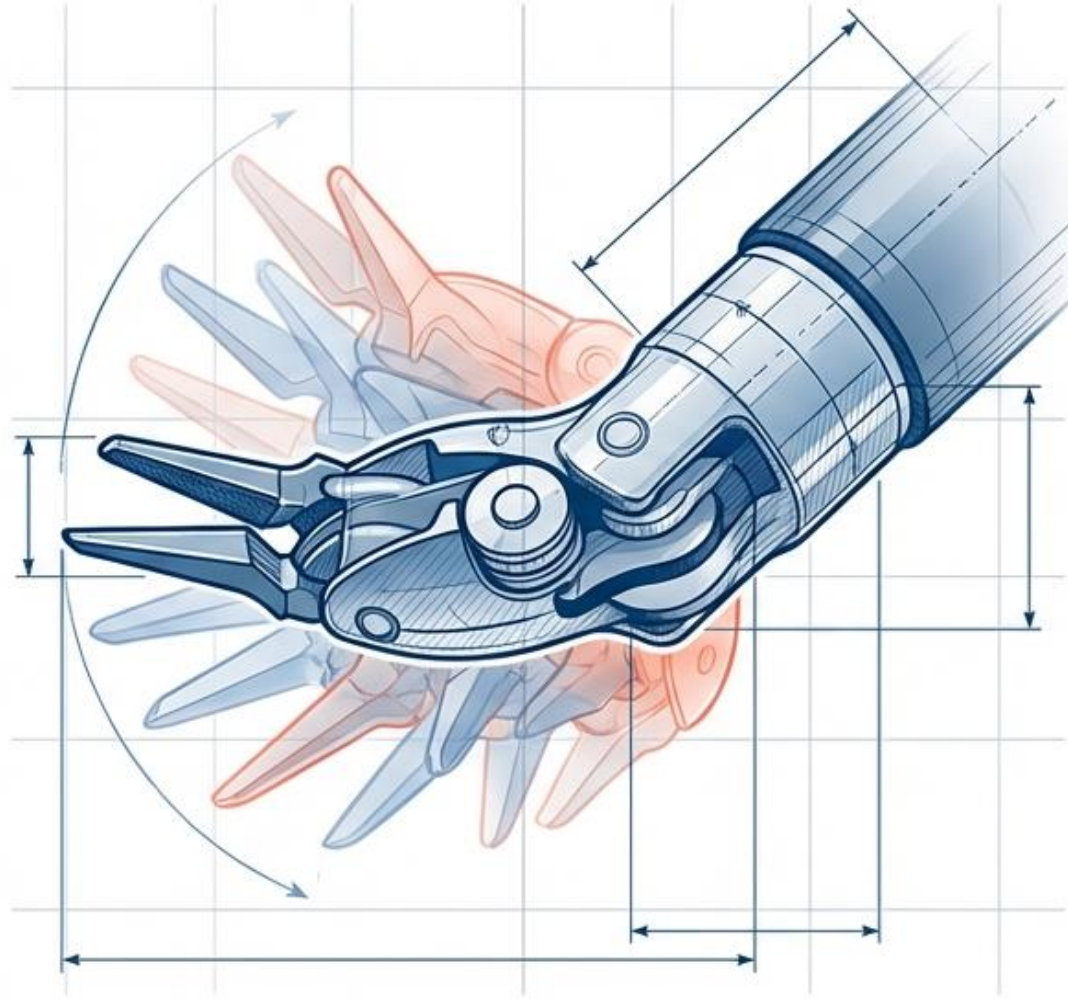


Why Do We Need Robots

- Robots are used where the work is too labor intensive, boring or dangerous for a human being.
- The three D's of Robotics: Robots perform tasks that are considered **dull, dirty and dangerous** for human beings.
- Robots have a better accuracy, repeatability and work efficiency than human workers.

Robots are (The Core Profile: Capabilities and Limitation)

- ✓ **Performance:**
Accurate, precise, and untiring.
- ✓ **Scale:**
Adaptable sizing (smaller or larger than humans).
- ✓ **Connectivity:**
Remote operation & instant information access.



Critical Limitation

Robots are not always able to **operate autonomously in highly complex, uncertain environments.**

Requirement:
Necessity of human interaction for decision-making.



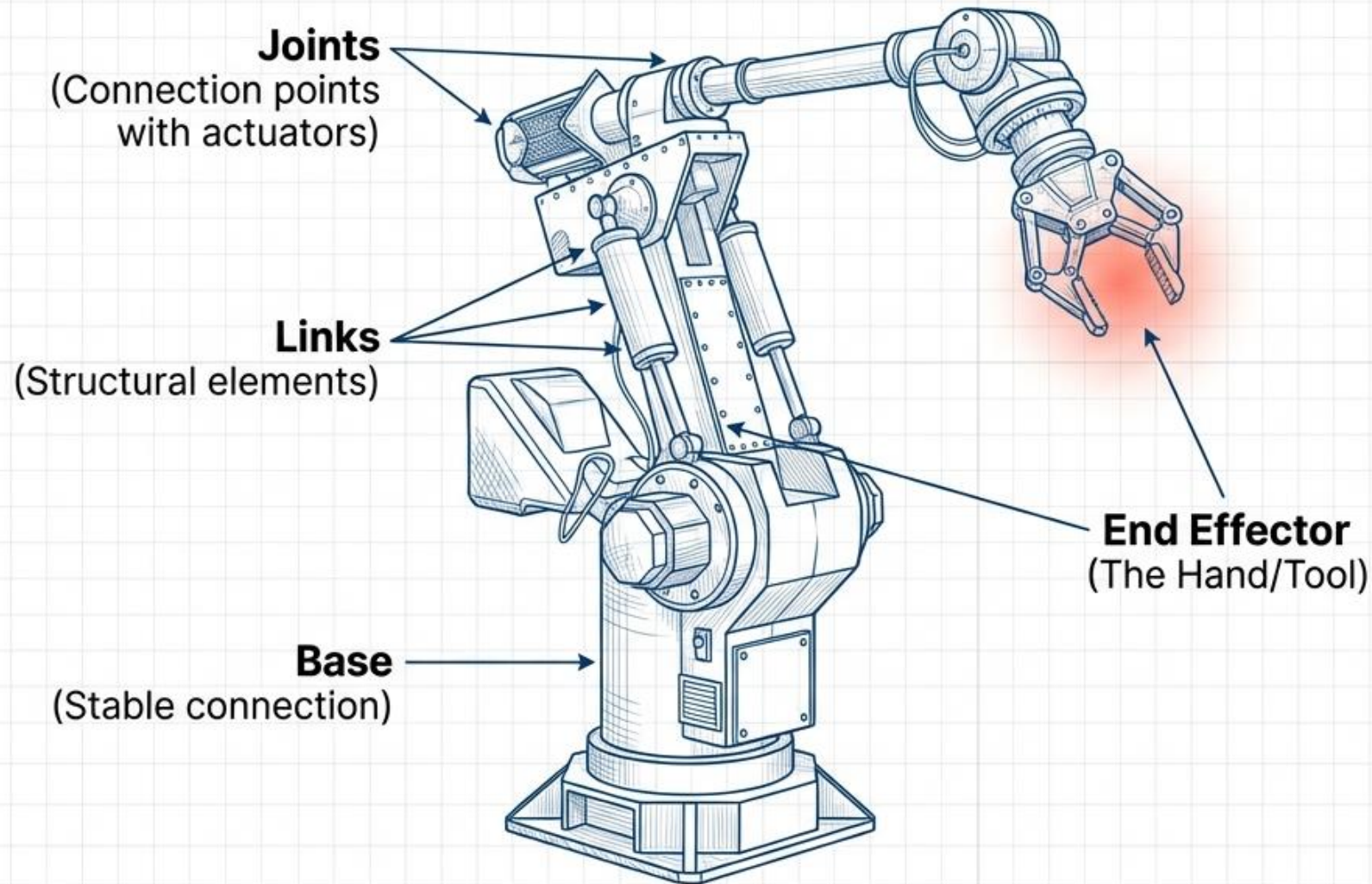
What is a Robot?

- A manipulator (or an industrial robot) is composed of a series of links connected to each other via joints. Each joint usually has an actuator (a motor for eg.) connected to it.
- These actuators are used to cause relative motion between successive links. One end of the manipulator is usually connected to a stable base and the other end is used to deploy a tool.





What is a Robot? (Anatomy I: The manipulator and end effector)



Manipulator/Rover: The main body composed of series of links connected via joints.

End Effector: The specific tool connected to the last joint, designed to interact with the environment.





Robot Components (Manipulator)

- ♦ **Manipulator or Rover: Main body of robot**
(Links, Joints, other structural element of the robot)
- ♦ **End Effector: The part that is connected to the last joint(hand) of a manipulator**
- ♦ **Actuators: Muscles of the manipulators**
(servomotor, stepper motor, pneumatic and hydraulic cylinder)
- ♦ **Sensors: To collect information about the internal state of the robot or To communicate with the outside environment**
- ♦ **Controller: Similar to cerebellum.**
It controls and coordinates the motion of the actuators.
- ♦ **Processor: The brain of the robot.**
It calculates the motions and the velocity of the robot's joints, etc.
- ♦ **Software: Operating system, robotic software and the collection of routines.**

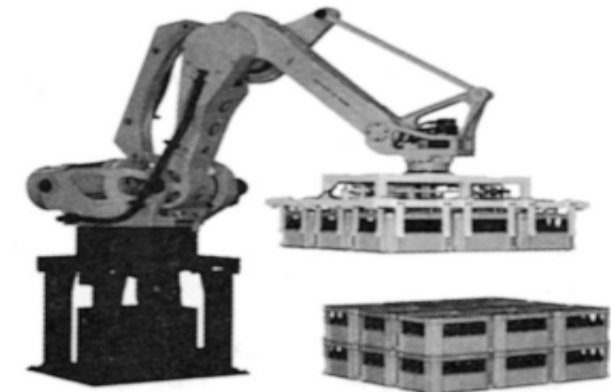
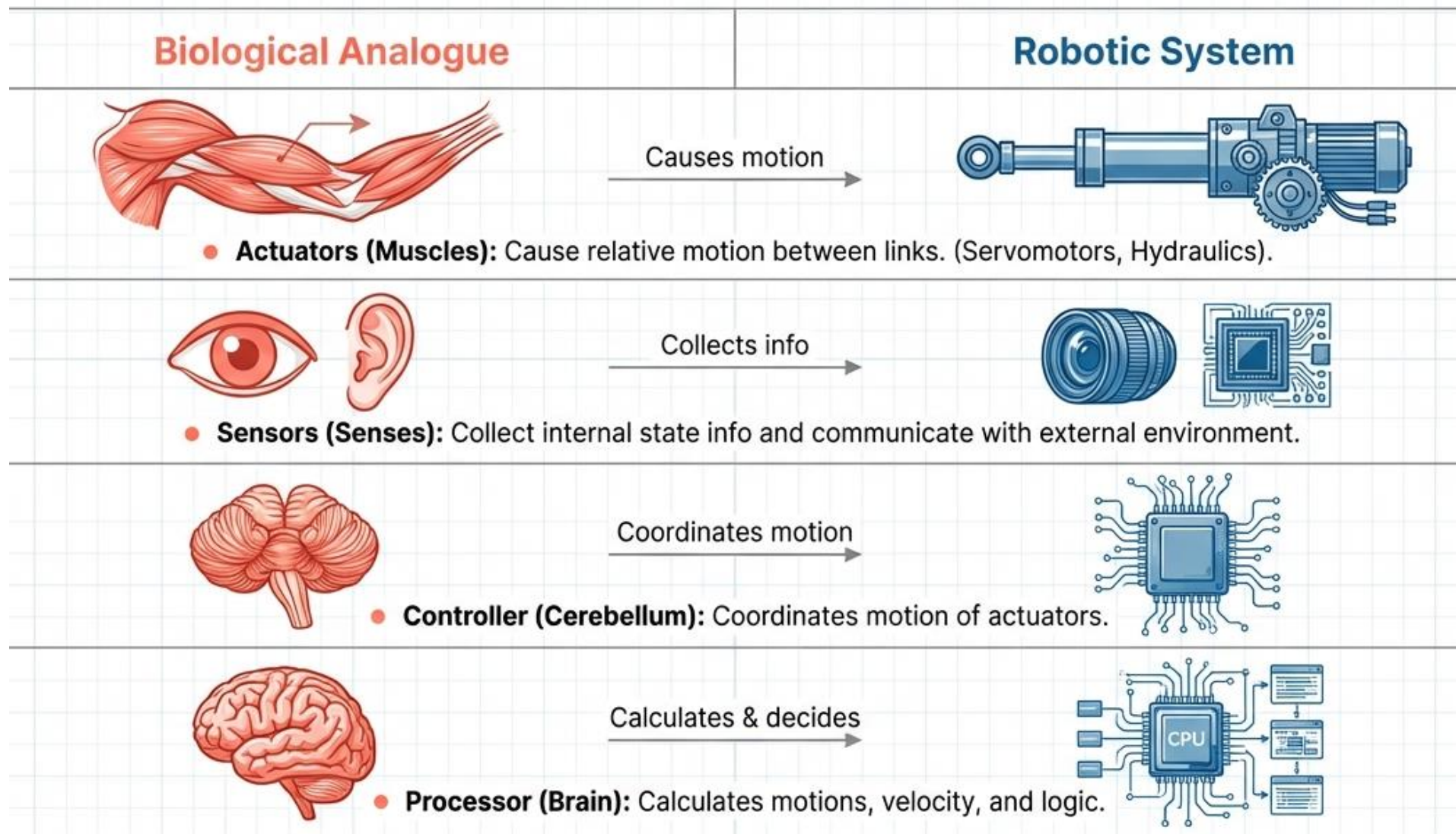
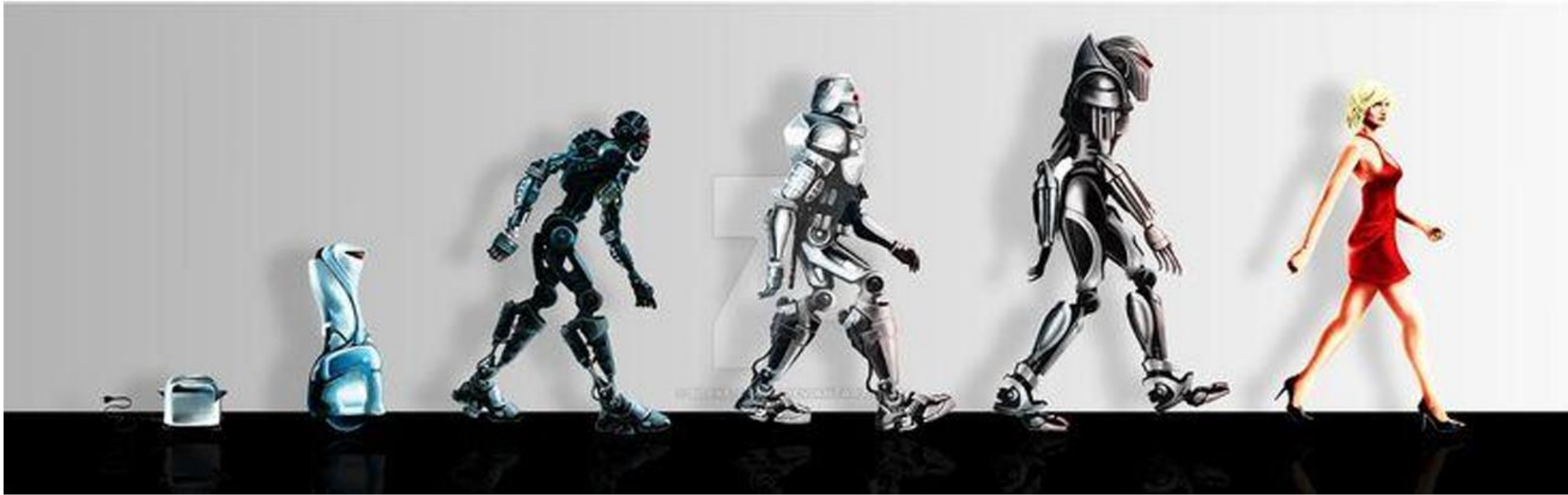


Fig. 1.2

What is a Robot? (Anatomy II: The Systems and Biological Analogues)



What is a Robot?



EVOLUTION OF THE CYLON





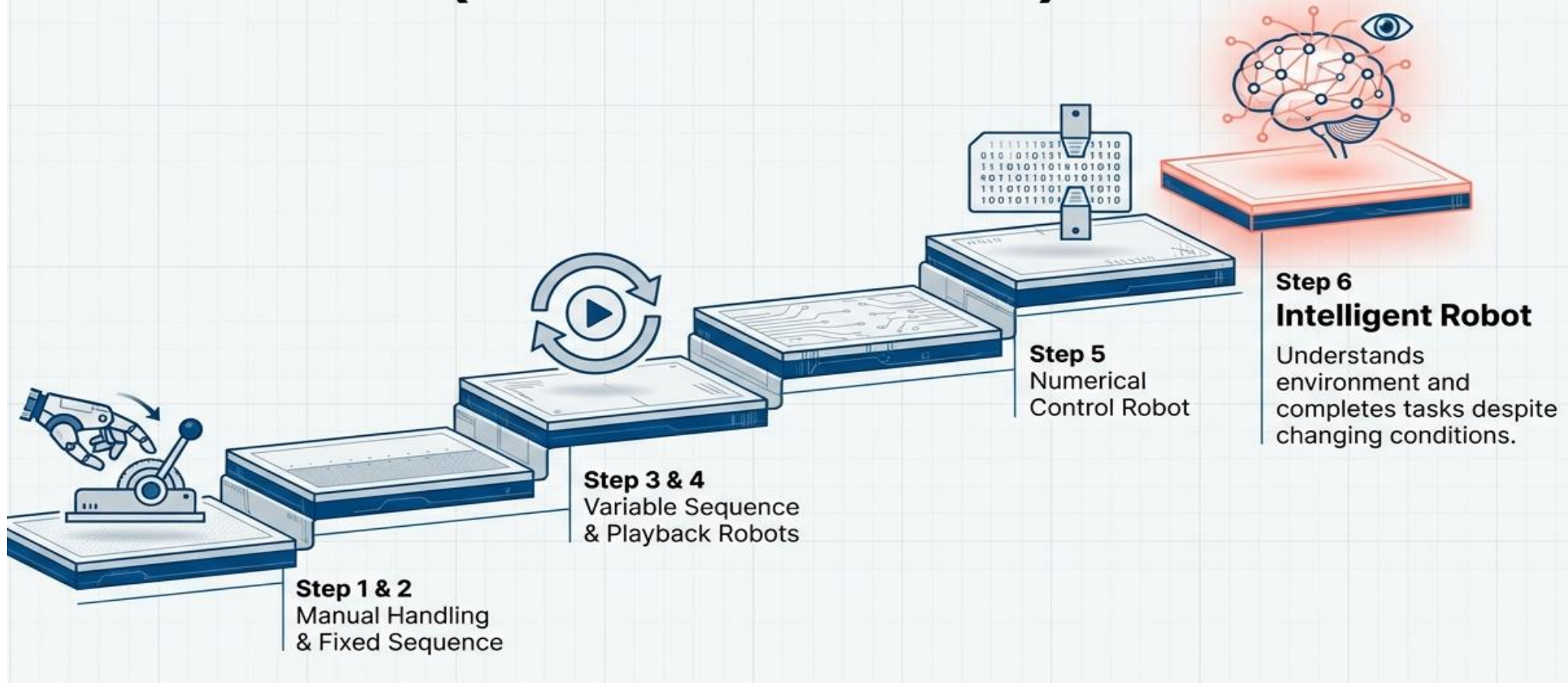
Classification of Robots

Japanese Industrial Robot Association (JIRA):

- **class 1: *manual handling devices***
 - device with several degrees of freedom actuated by operator
- **class 2: *fixed sequence robot***
 - handling device which performs the successive stages of a task according to a predetermined, unchanging method, which is difficult to modify
- **class 3: *variable sequence robot***
 - as class 2, but the stages can be easily modified
- **class 4: *playback robot***
 - the robot can repeat (playback) a sequence of tasks recorded from a human operator leading or controlling the robot
- **class 5: *numerical control robot***
 - human operator supplies the robot with a movement program
- **class 6: *intelligent robot***
 - a robot with the means to understand its environment and the ability to complete tasks despite changes in conditions

Classification of Robots

The Ladder of Complexity (JIRA Classification)



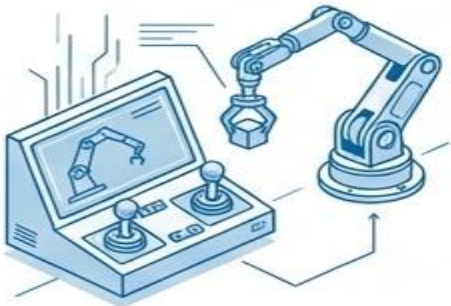
Classification of Robots

Association Française de Robotique (AFR)

- **Type A:** Manual Handling Devices/telerobotics
- **Type B:** Automatic Handling Devices/predetermined cycles
- **Type C:** Programmable, Servo controlled robot, continuous point-to-point trajectories
- **Type D:** Same type with C, but it can acquire information.

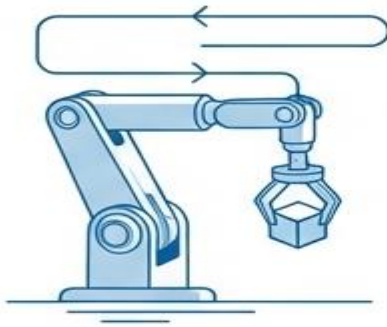
Type A: Manual Handling

Tele-robotics



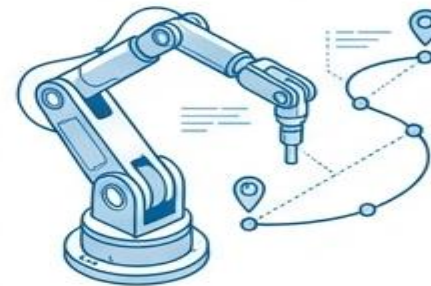
Type B: Automatic Handling

Predetermined cycles



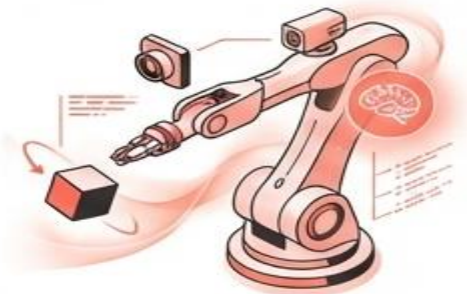
Type C: Programmable

Servo-controlled,
point-to-point
trajectories



Type D: Advanced Interaction

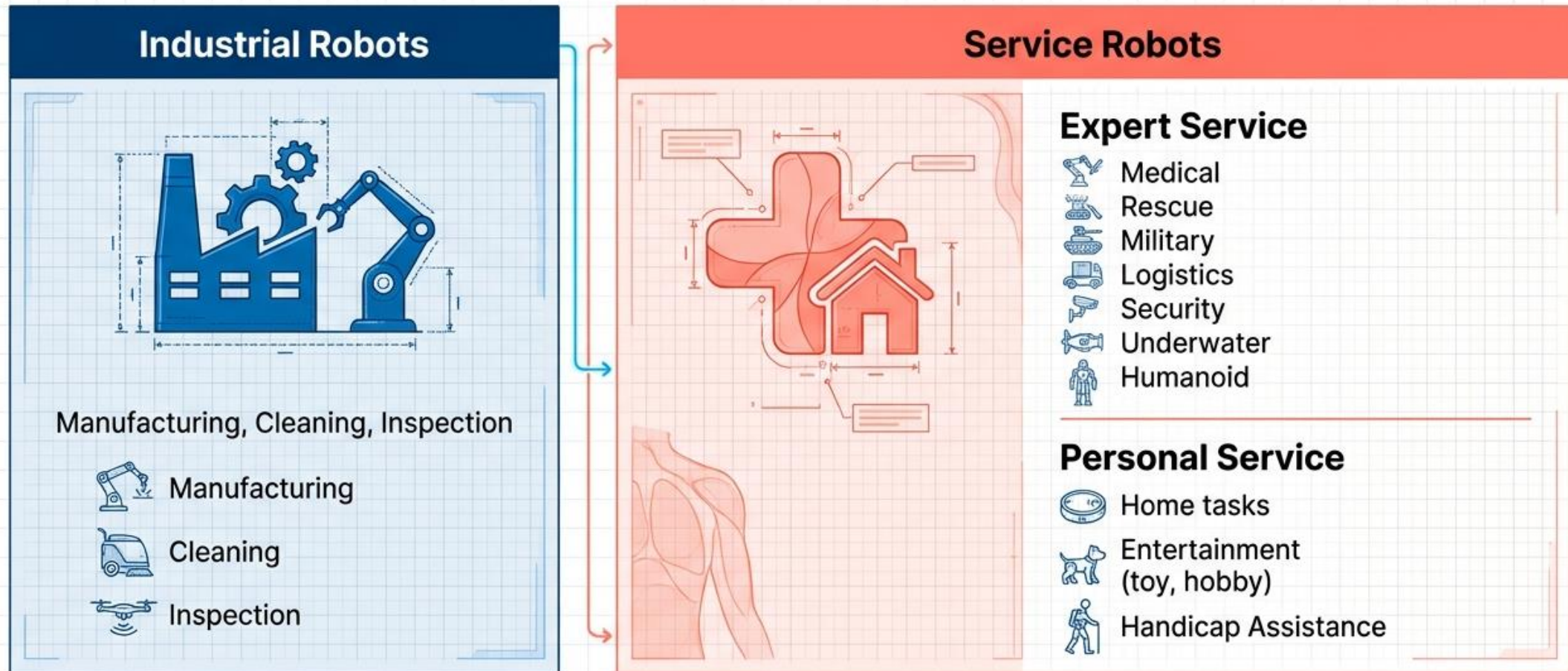
Same as Type C, but
capable of acquiring
information from the
environment



Classification of Robots

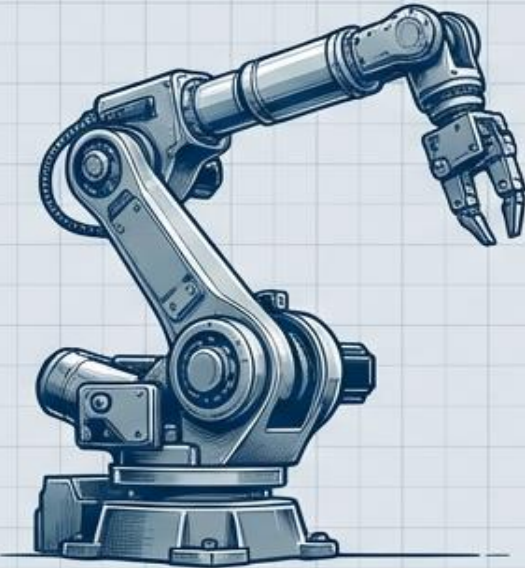
Global Classification by Application (IFR)

International Federation of Robotics Framework



Robotics (Brief History): Paradigm shift from substitution to Coexistence

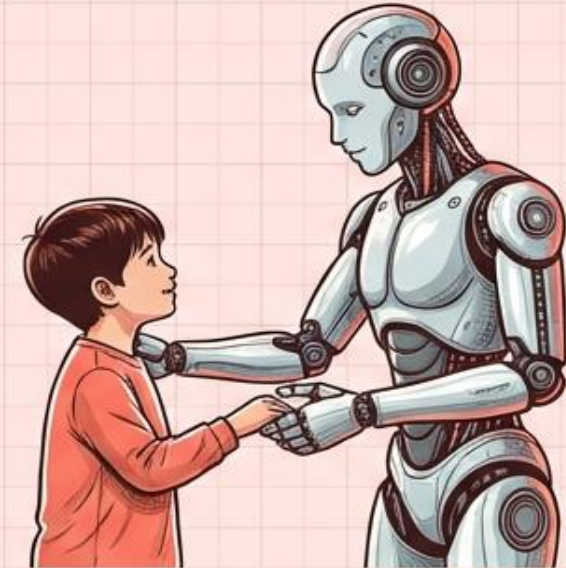
Classic Robot (1990 - 2005)



- Substitution of labor
- Repetitive work
- Goal: Productivity

Catalyst:
Technology Innovation
& Changing
Societal Needs

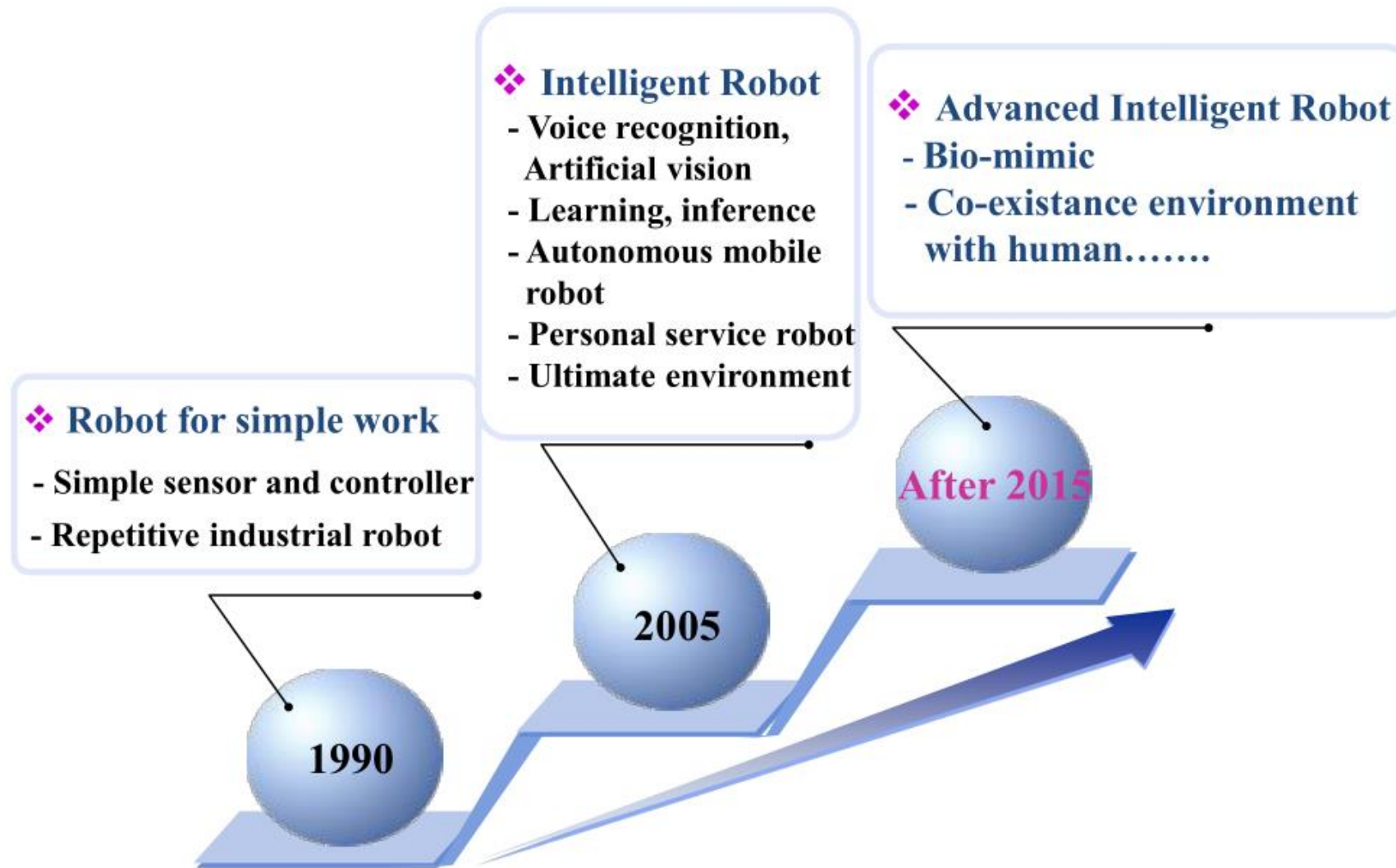
Intelligent Robot (Post-2015)



- Coexistence with humans
- Self-motion & Bio-mimicry
- Goal: Quality of Life

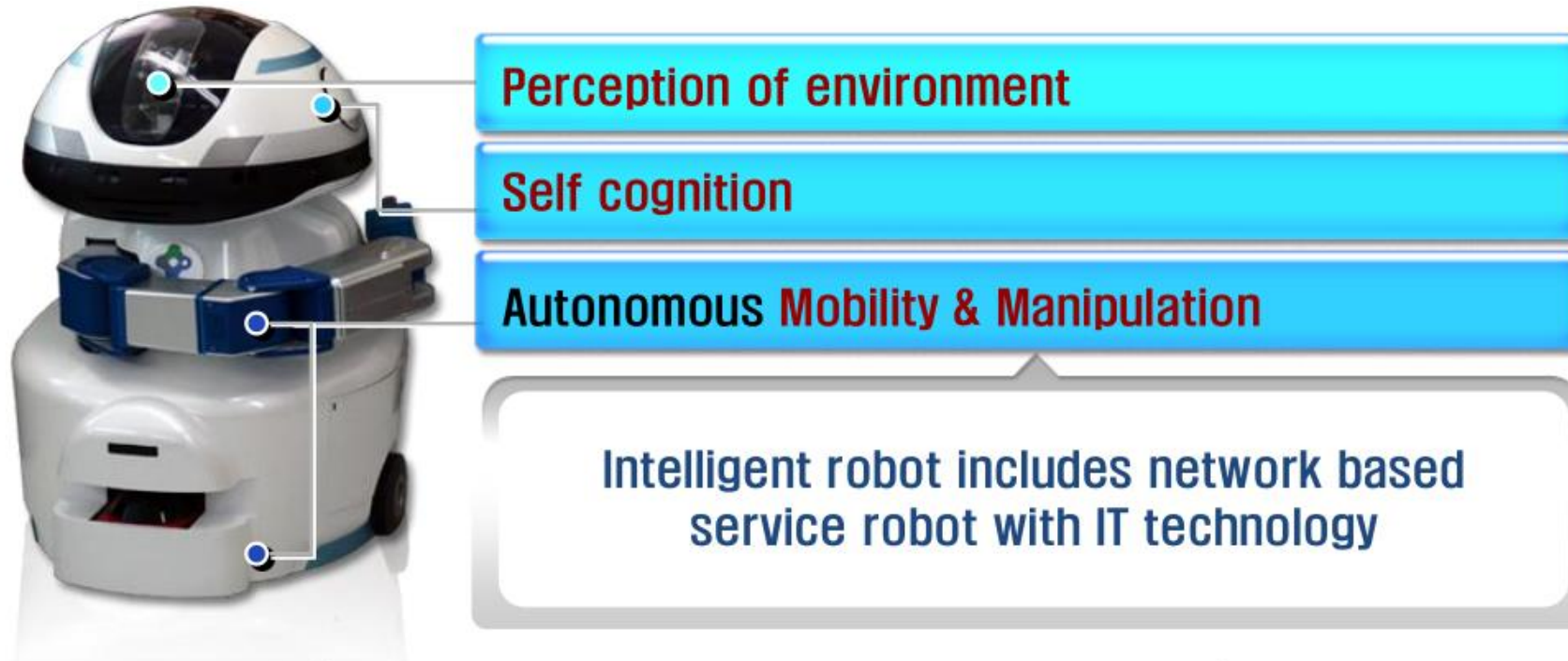


Developing to Intelligent Robot





Definition of Intelligent Robot and Classification



Classification	Service Robot		Industrial Robot
	Service Robot for Personal Use	Service Robot for Professional Use	

Source : IFR (International Federation of Robot)



Classification of Intelligent Robot

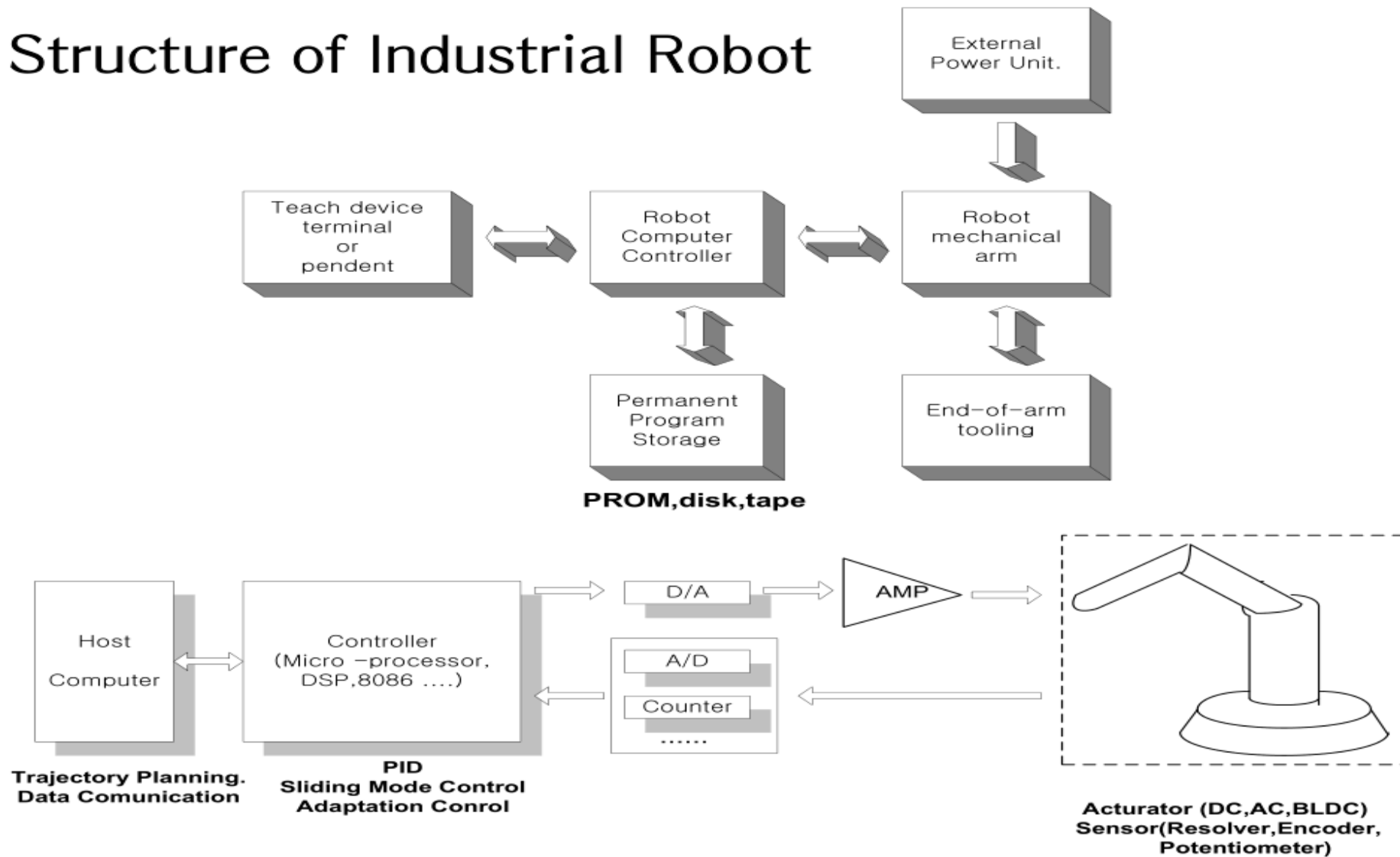
IFR(International Federation of Robotics) Robot Classification

Classification of Robot	Specific Classification by Application Areas
Industrial Robot	Manufacturing
Expert Service Robot	Field, Cleaning, Inspection Construction, Logistics, Medical, Military, Rescue, Security, Underwater, Humanoid
Personal Service Robot	Home tasks, Entertainment (toy, hobby), Handicap Assistance



Structure of Industrial Robot

Structure of Industrial Robot



Robotics (Brief History)

First Generation (Manufacturing)



Descendants of Unimate. Blind automation for repetitive tasks.

Next Generation (Vision & Awareness)



Built-in vision capability and sensory awareness.



Robotics (Manipulator)



Robotics (Brief History)



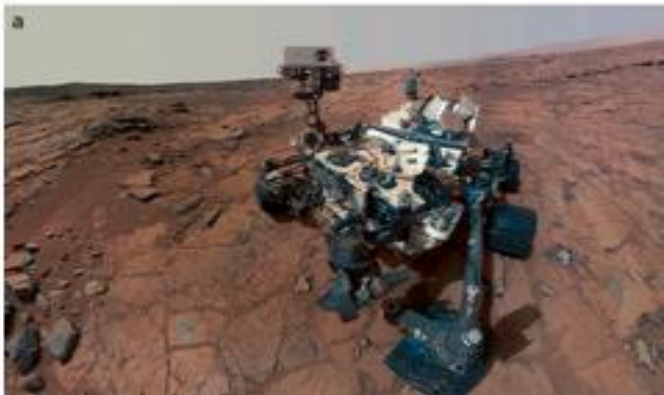
Small autonomous underwater vehicle
(Todd Walsh © 2013 MBARI)



self driving car (image courtesy Dept. Information Engineering, Oxford Univ.)



Honda's Asimo humanoid robot (image courtesy Honda Motor Co. Japan)



Mars Science Lander, Curiosity, self portrait taken at "John Klein".



Cheetah legged robot (image courtesy Boston Dynamics)



Savioke Relay delivery robot (image courtesy Savioke)

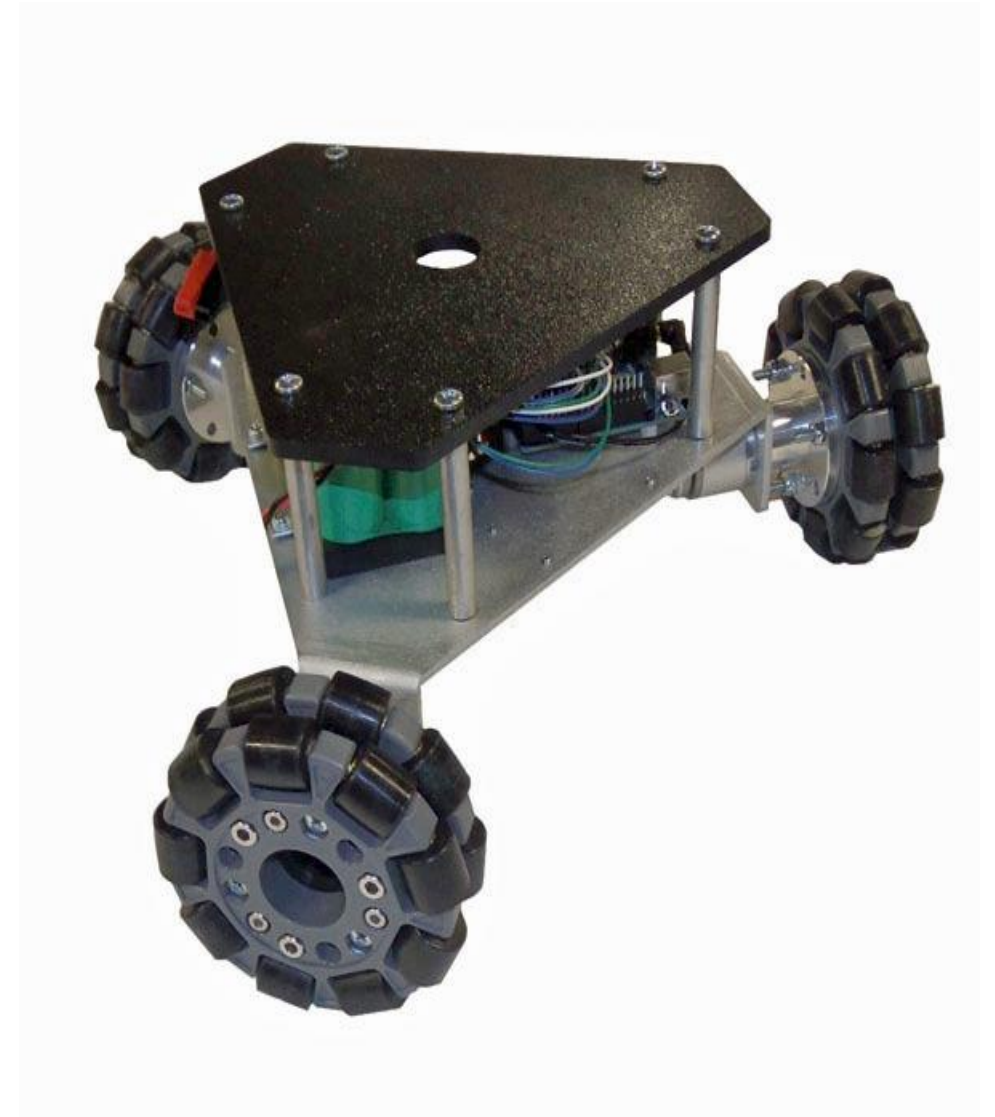


Robotics (Manipulator)





Robotics (Wheeled Robot)





Robotics (Legged Robot)





Robotics (Humanoid Robot)





Robotics (Unmanned Aerial Vehicle)



Robotics (Autonomous Underwater Vehicle)



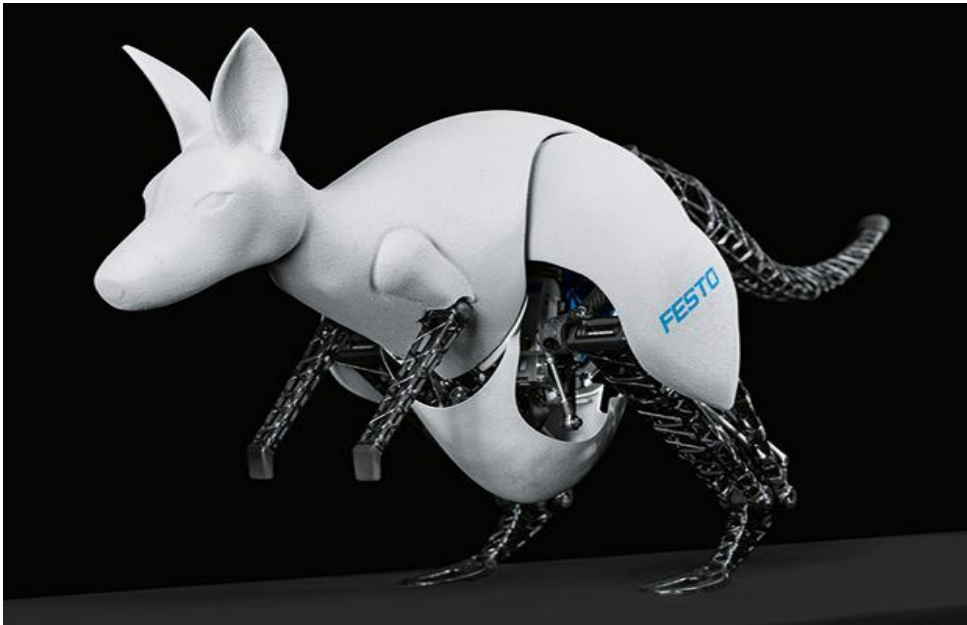
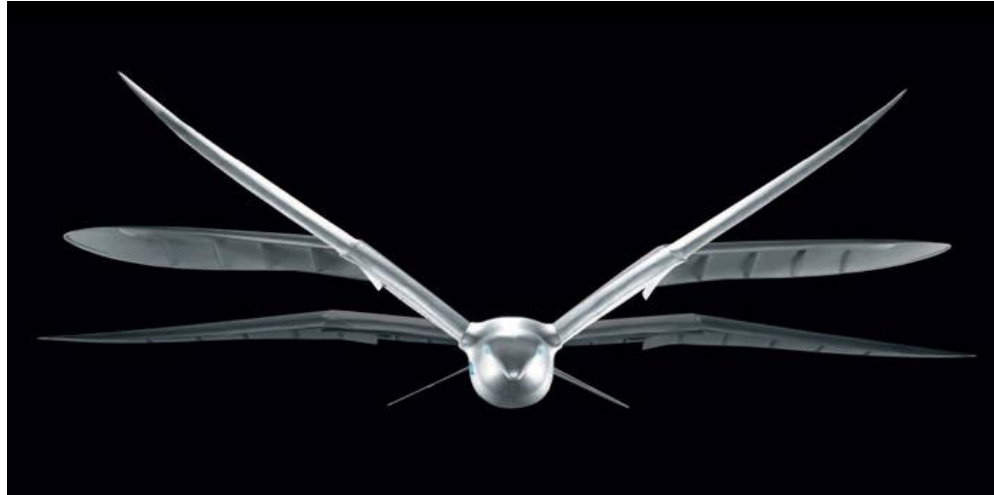
Autonomous Underwater vehicle



Global Hawk unmanned aerial vehicle
(photo courtesy of NASA)



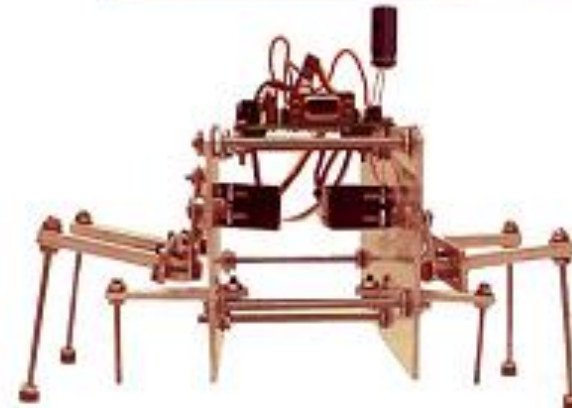
Robotics (Bio-inspired Robot)





Robotics

Robotics



Smart Robotics Application

BigDog



System Can

- Carry 340 lb
- Run 4 mph
- Climb, run, and walk
- Move over rough terrain

Advantages

- Robot with **rough-terrain mobility** that could carry equipment to remote location.



Smart Robotics Applications

- Robots can vacuum floors and clean gutters **so you don't have to.**

Cleans Gutter



Vacuum Floors



MECH-

Level of Human Input Varies

Oral



Manual



Cooperative manipulation



Teleoperation



Autonomous





Classification of Medical Robotics

❖ **Surgical Robotics** (<https://www.youtube.com/watch?v=JJK4fZX48bQ>)

- **Minimally Invasive Surgery (MIS) Robots** (e.g., da Vinci Surgical System)
- **Orthopedic Surgery Robots** (e.g., MAKOplasty for joint replacement)
- **Neurosurgical Robots** (e.g., ROSA, Neuromate)
- **Microsurgery Robots** (e.g., eye, ENT, vascular procedures)



❖ **Rehabilitation Robotics** (<https://www.youtube.com/shorts/d1Tc5LYQeQA>)

- **Exoskeletons** (for stroke and spinal cord injury recovery)
- **End-Effector Based Rehabilitation Devices** (robotic arms/legs for repetitive therapy)
- **Balance and Gait Training Robots**



Classification of Medical Robotics

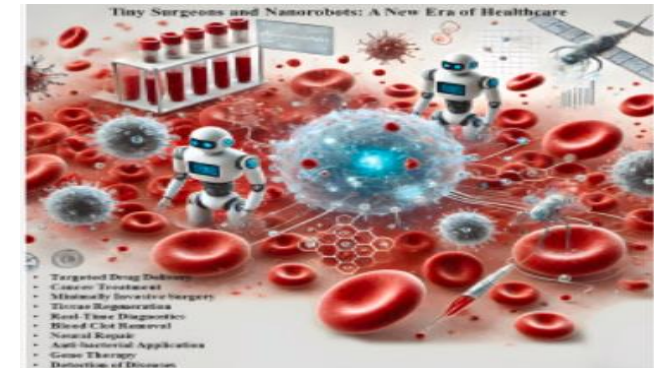
❖ **Tele-Robotics** (https://www.youtube.com/watch?v=R-sP_pju81E)

- **Remote Surgery Robots** (telesurgery, remote operation)
- **Telepresence Robots** (patient monitoring, consultations)



❖ **Nano-Robotics** (<https://www.youtube.com/watch?v=YO6oQ1xan8I>)

- **Drug Delivery Nanobots** (targeted therapy, e.g., cancer treatment)
- **Cellular/Nano-manipulation Robots** (gene and cell therapy)
- **Diagnostic Nanorobots** (early disease detection)



❖ **Diagnostic & Imaging Robotics** (<https://www.youtube.com/watch?v=A18FpBENePg&t=74s>)

- **Biopsy Robots** (precise tissue sampling)
- **Imaging Assistance Robots** (MRI, CT-guided interventions)
- **Ultrasound Robots**





Classification of Medical Robotics

- ❖ **Assistive Robotics** (<https://www.youtube.com/watch?v=gFA4O8I3fg8>)
 - **Robots for Elderly Care** (mobility support, daily assistance)
 - **Companion Robots** (mental health and social interaction support)
 - **Feeding & Personal Care Robots**



- ❖ **Pharmacy & Hospital Automation**
(<https://www.youtube.com/shorts/A8QQGCic0aA>)
(<https://www.youtube.com/shorts/lNkqkVceIA4>)

- **Drug Dispensing Robots**
- **Logistics Robots** (medication, blood, and sample transport)
- **Sterilization & Disinfection Robots**



Why Medical Robot?

➤ Improvement of Surgical Accuracy

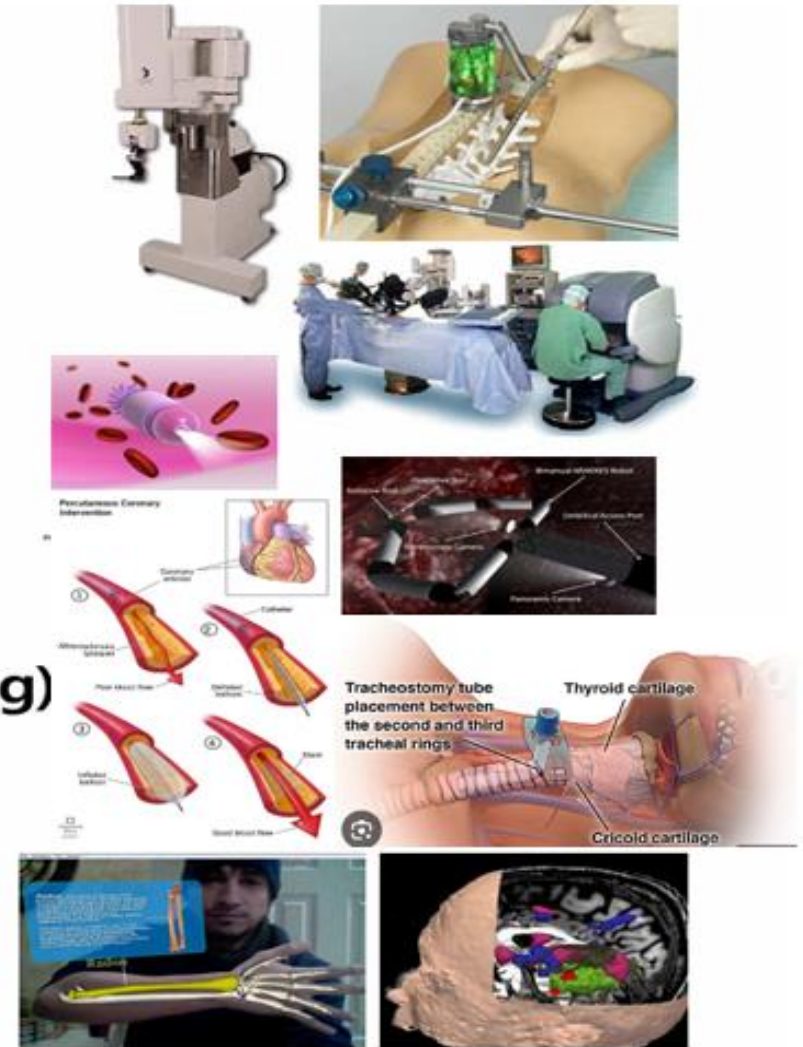
- Bone-machining CAD/CAM system
- Surgical tool guidance for screwing/biopsy

➤ Minimal Invasiveness

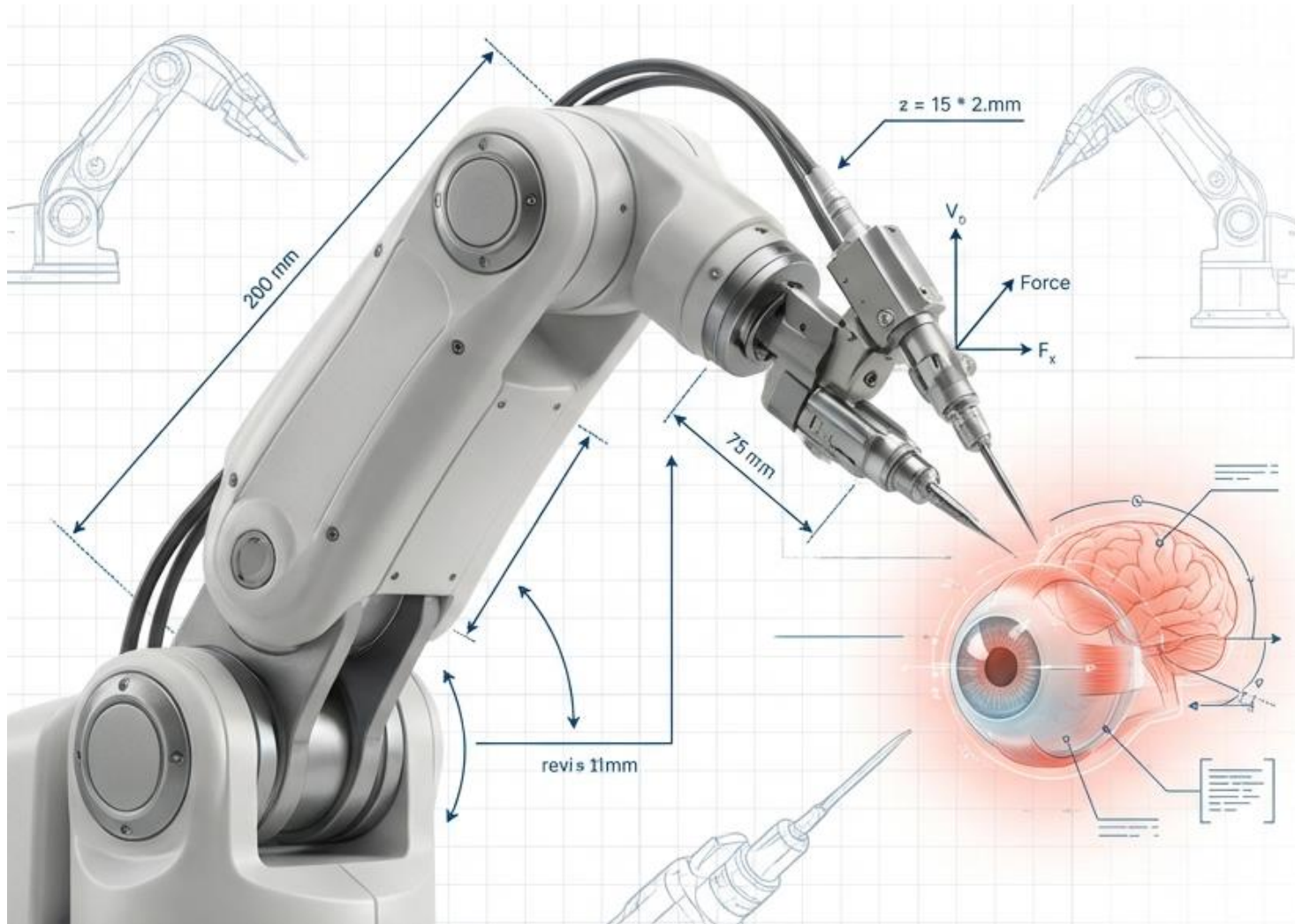
- Laparoscopic (endoscopic) surgery robot
- Percutaneous intervention system

➤ Richer Information (Various Medical Imaging)

- Medical Images i.e. CT, MRI, Angiography...
- MRI-compatible surgical robots
- Augmented Display



Medical Applications (Surgical Robotics)

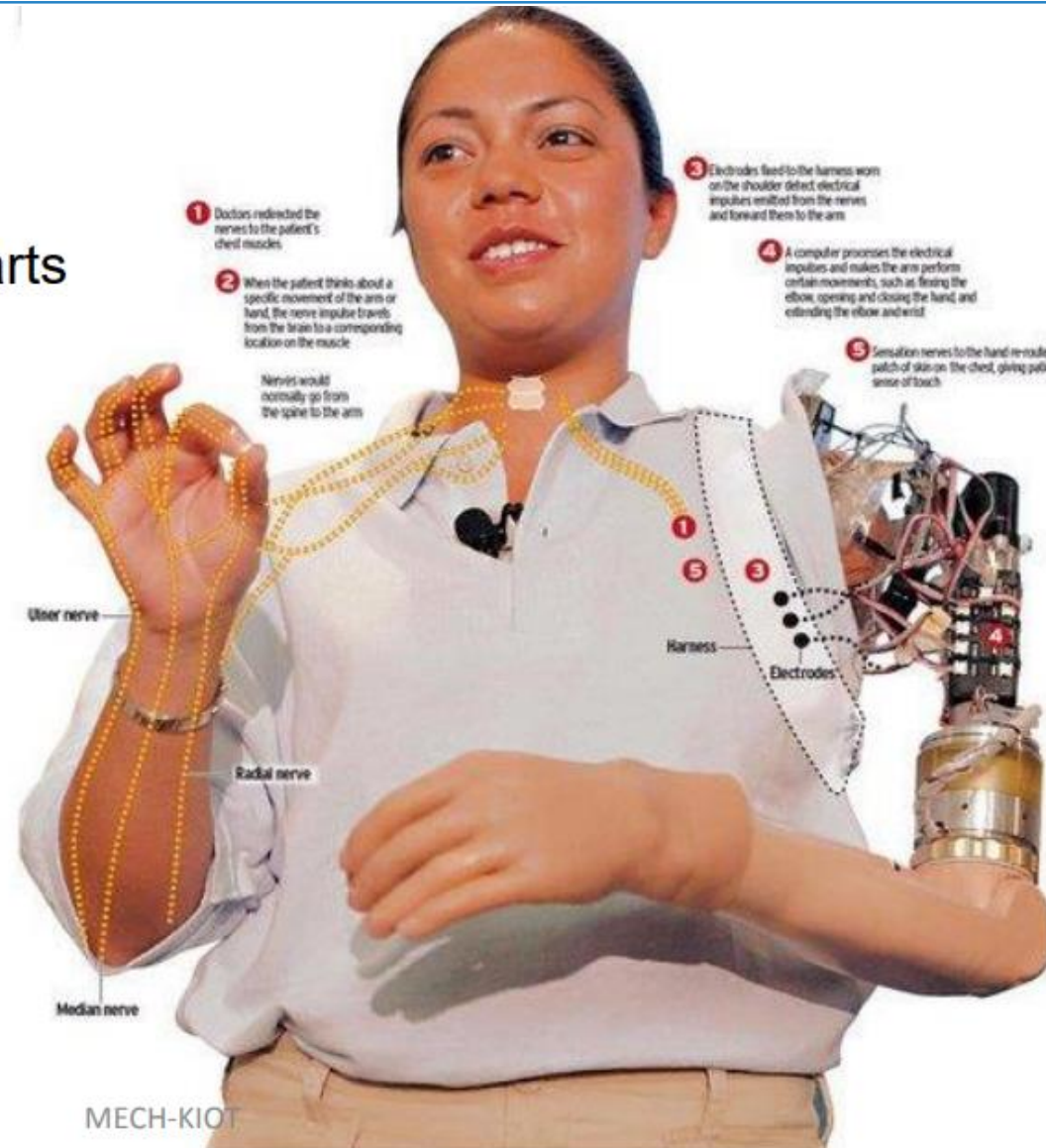


	Minimally Invasive Surgery (MIS): da Vinci Systems allowing complex procedures with tiny incisions.
	Specialized Disciplines: Orthopedic (MAKOplasty), Neurosurgical (ROSA), and Microsurgery (Eye/ENT).
	Remote Capability: Telesurgery and Telepresence for distance monitoring.

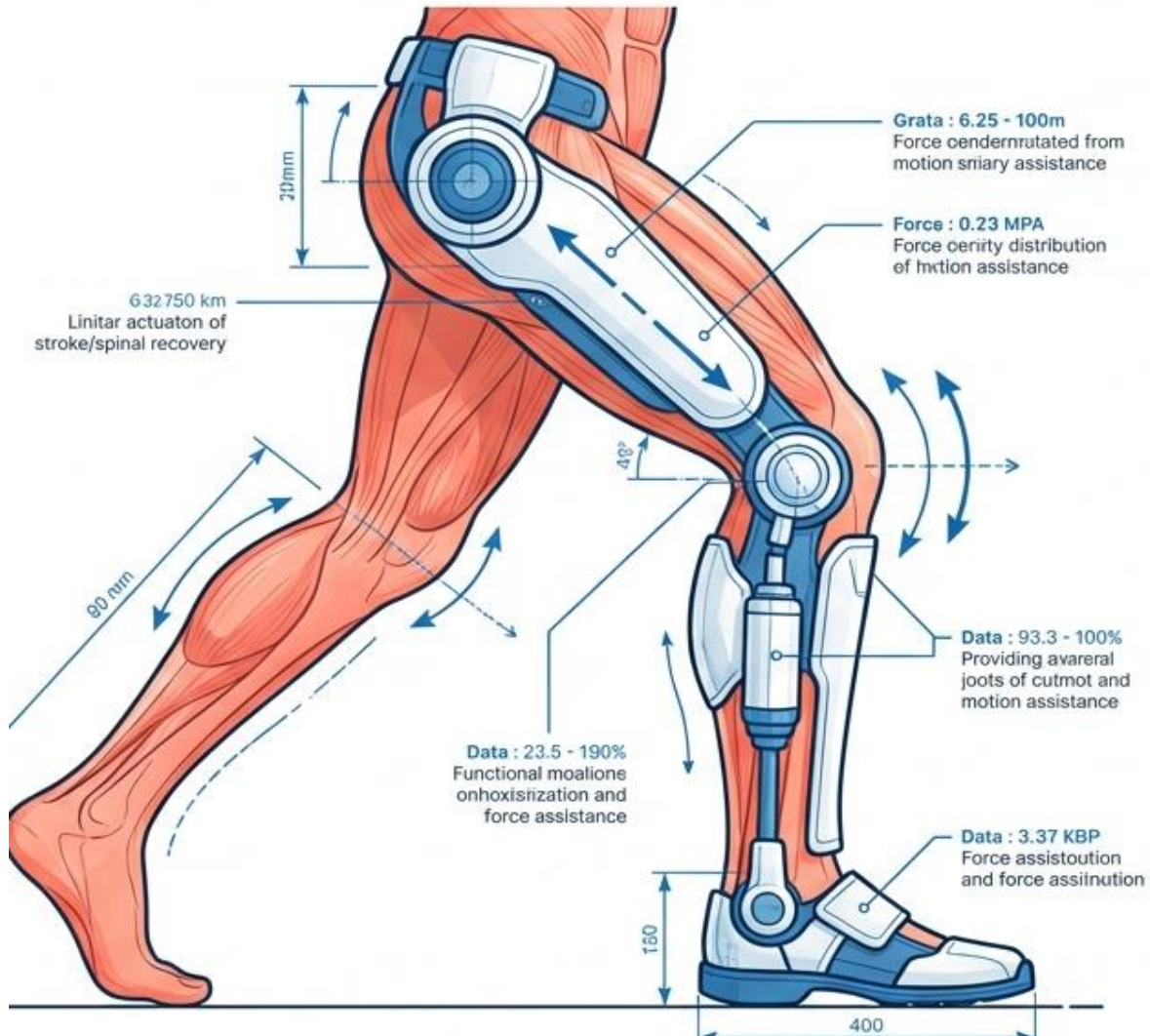
Medical Applications

Prosthetics

- Arms, Legs, and other body parts can be replaced with electromechanical ones.



Medical Applications (Rehabilitation and Assistive Care)



Physical Rehabilitation:
Exoskeletons for stroke/spinal injury recovery, and robotic limbs for repetitive therapy.

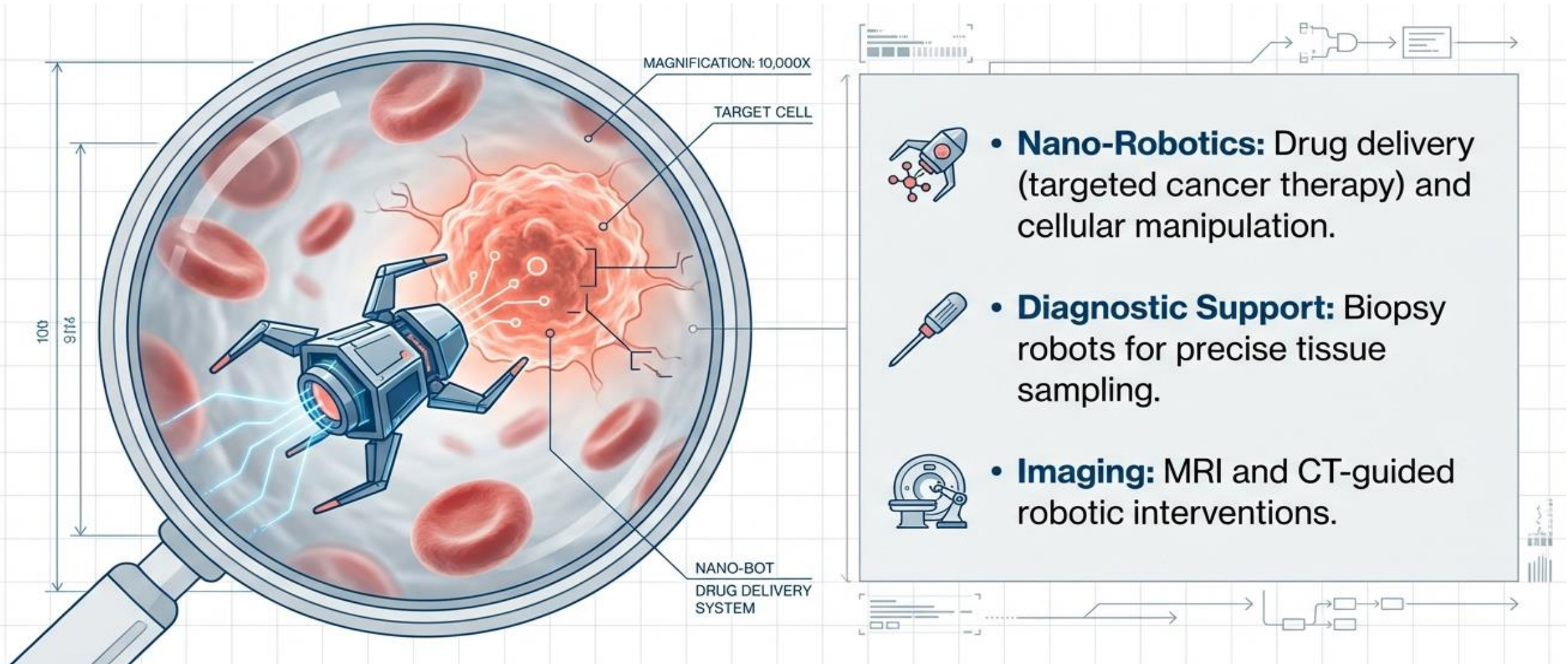


Social & Daily Assistance:
Mobility support bots for elderly care.

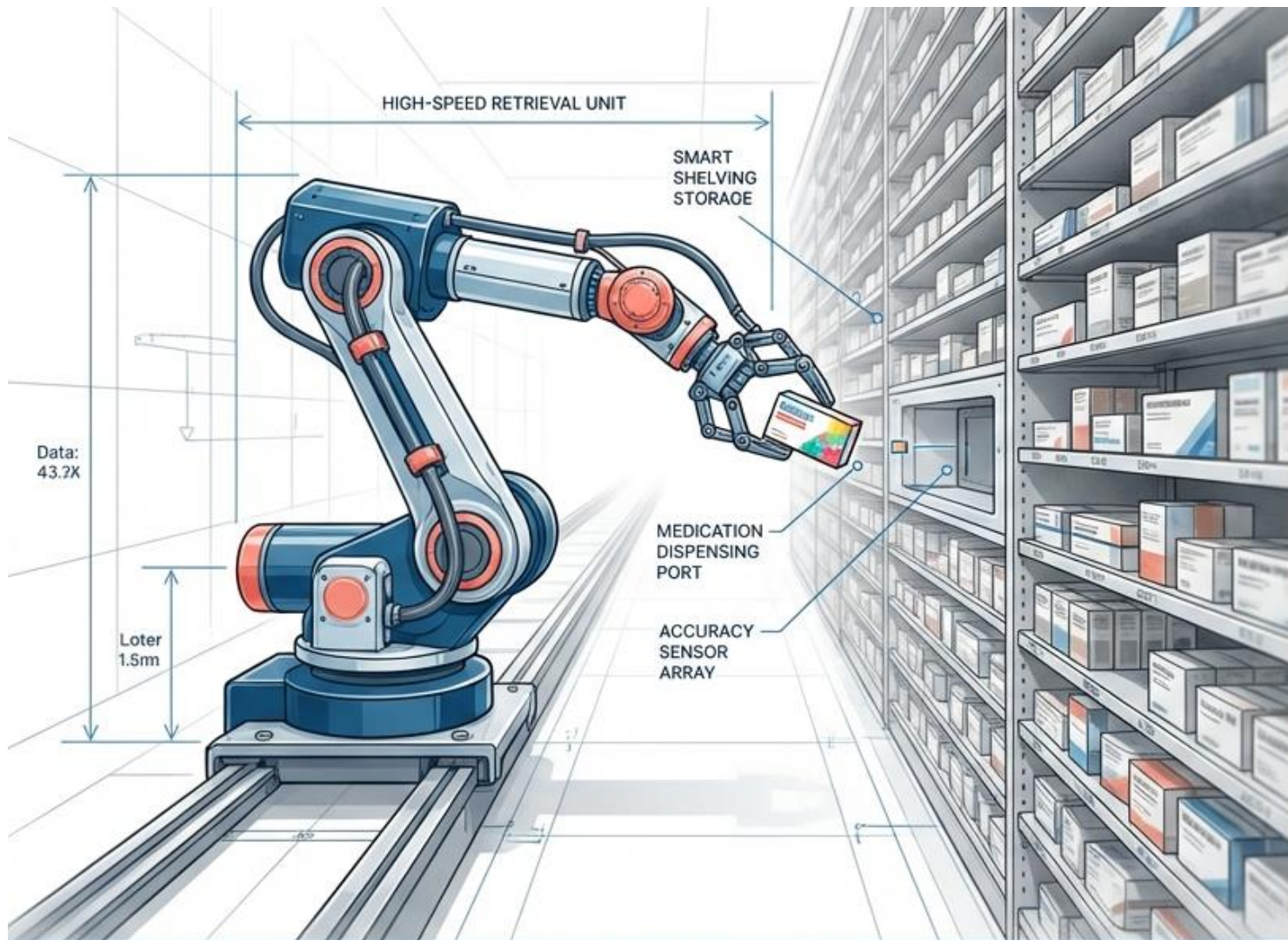


Companion Robots:
Providing social interaction and mental health support.

Medical Applications (Micro-Scale: Nano Robotics and Diagnostic)



Medical Applications (Automated Hospital Ecosystem)



Pharmacy Automation

Automated dispensing for dosage accuracy.



Logistics

Autonomous transport of blood, samples, and medication.

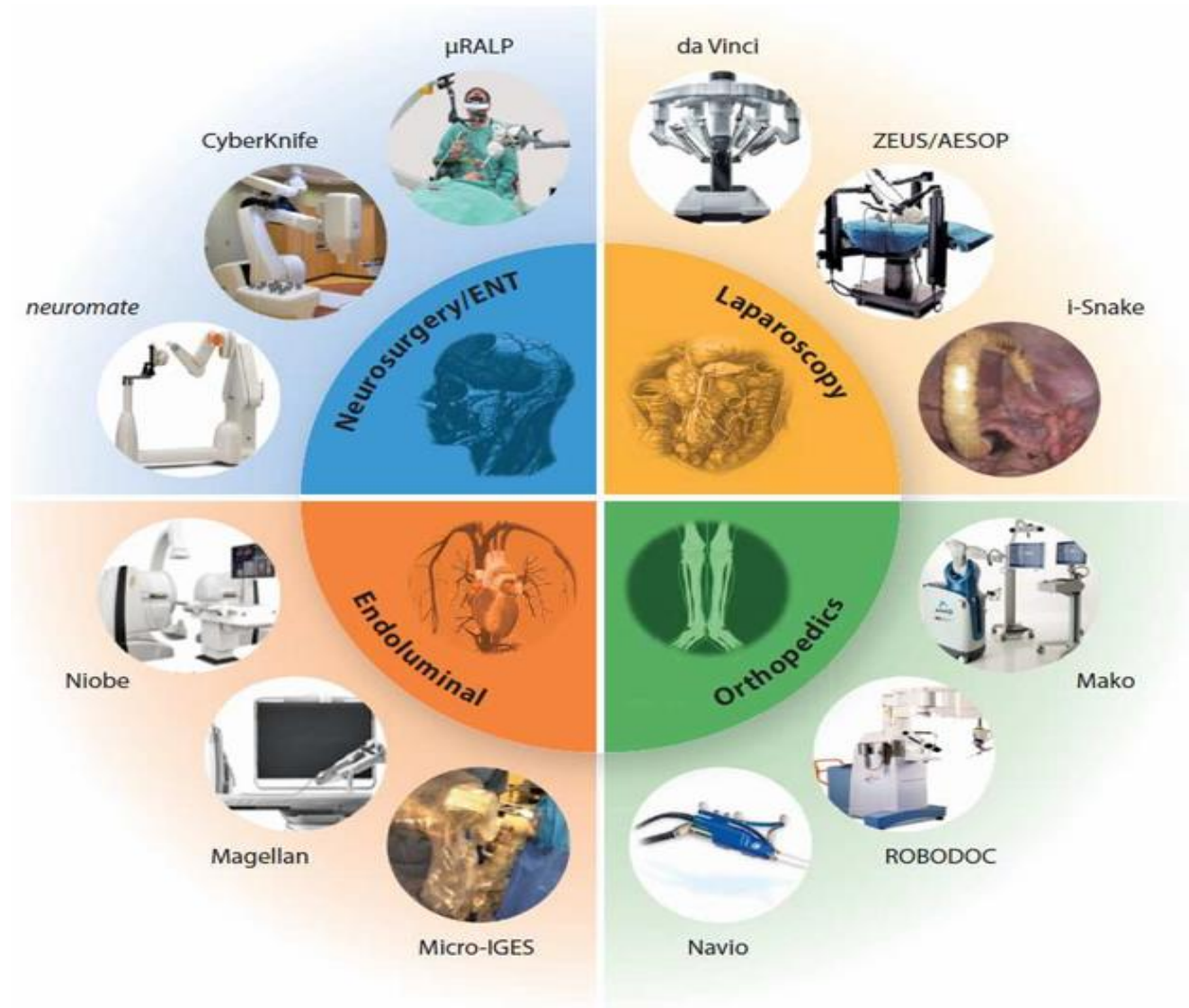


Sanitation

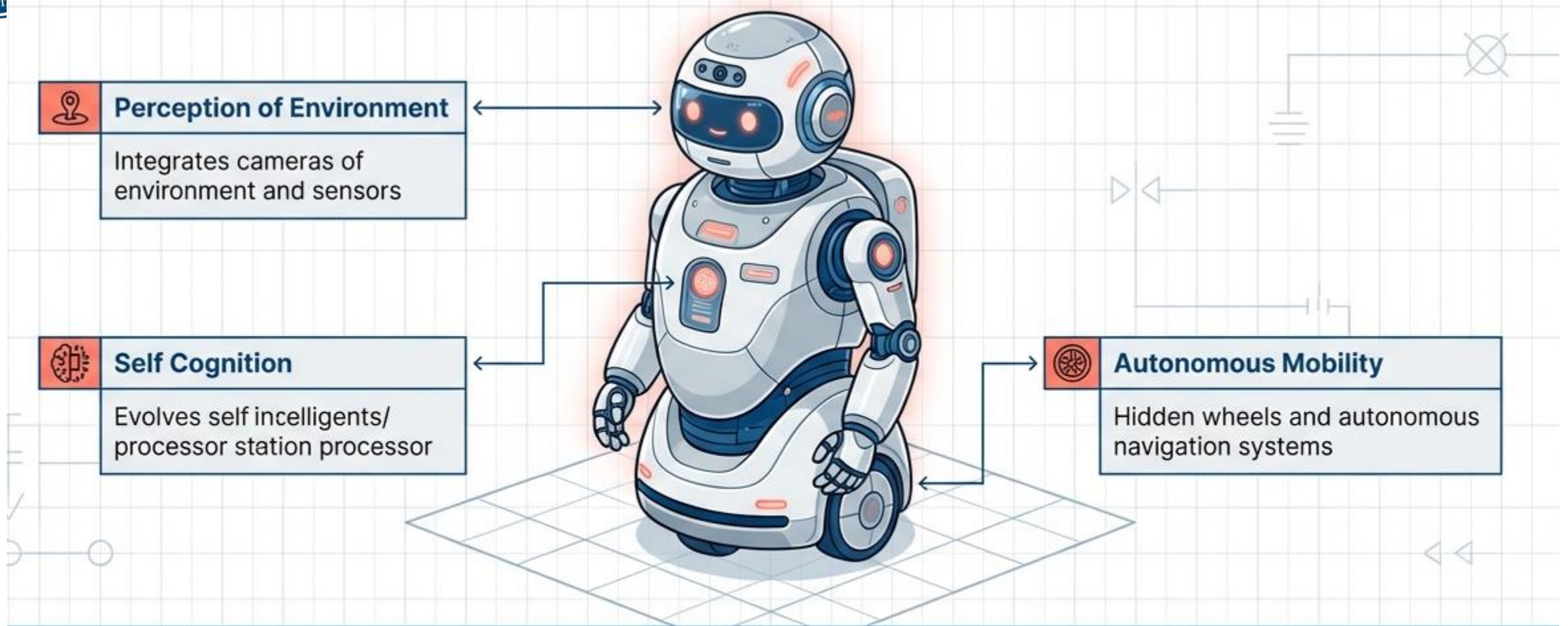
UV/Chemical sterilization robots maintaining safe environments.



Examples of Surgical Robot Specializations



Summery: The Trajectory of Intelligence



We have evolved from fixed-sequence tools to intelligent partners that coexist with humans and save lives.



Advantages VS. Disadvantages of Robots

- Robots increase productivity, safety, efficiency, quality, and consistency of products.
- Robots can work in hazardous environments without the need.
- Robots need no environmental comfort.
- Robots work continuously without experiencing fatigue of problem.
- Robots have repeatable precision at all times.
- Robots can be much more accurate than human.
- Robots replace human workers creating economic problems.
- Robots can process multiple stimuli or tasks simultaneously.
- Robots lack capability to respond in emergencies.
- Robots, although superior in certain senses, have limited capabilities in Degree of freedom, Dexterity, Sensors, Vision system, real time response.
- Robots are costly, due to Initial cost of equipment, Installation costs, Need for Peripherals, Need for training, Need for programming.



Ethical Issues

“robots taking jobs from people”

“whom to blame if a robotic car crashes into a human or gets into an accident”

“human vs robotic surgery”

“Should we use robots to look after our children, and even teach them?”

Any Questions?

